STAT A411/611 Class Notes

Instructor: Rieken Venema **Office:** SSB 158E
Office hours: MWF 12:00–12:45 **Class:** 2:30–3:45

Plan and Review

Today:	Sections 4.2 and 4.3.
Next Monday:	Sections 4.3 and 4.4.
Chapter 4:	Sections 4.1–4.6; continuous random variables.
Chapter 3:	Discrete random variables and their probability distributions.

Section 4.1 - Continuous Random Variables

A continuous random variable X is described by a **probability density function** (pdf) $p(x)$. Probabilities are computed as areas beneath this density.

Probability Densities and Cumulative Probabilities

A valid pdf satisfies

$$\boxed{p(x) \geq 0 \quad \forall x \in \mathbb{R}}, \qquad \boxed{\int_{-\infty}^{\infty} p(x)\,dx = 1}.$$

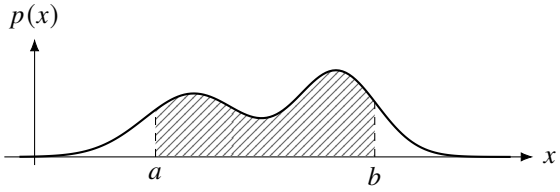
For any $a \leq b$,

$$\boxed{P(a \leq X \leq b) = \int_a^b p(x)\,dx}.$$

The cumulative distribution function (cdf) and its derivative are

$$\boxed{F(x) = P(X \leq x) = \int_{-\infty}^{x} p(t)\,dt}, \qquad \boxed{F'(x) = p(x)}.$$

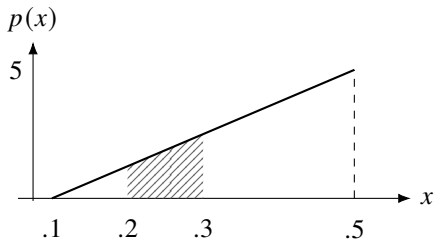
The derivative identity applies wherever p is continuous (and almost everywhere in general).



Lead-Concentration and Uniform-Distribution Review

For the lead-concentration density $p(x) = 12.5x - 1.25$ on $0.1 \leq x \leq 0.5$,

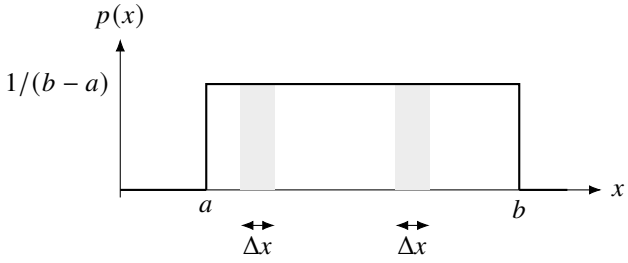
$$\boxed{P(0.2 \leq X \leq 0.3) = \int_{0.2}^{0.3} (12.5x - 1.25)\,dx = F(0.3) - F(0.2)}.$$



For a continuous uniform random variable $X \sim U(a, b)$, where $a < b$,

$$\boxed{p(x) = \begin{cases} \dfrac{1}{b-a}, & a \le x \le b, \\ 0, & \text{otherwise}, \end{cases}} \qquad \boxed{F(x) = \begin{cases} 0, & x < a, \\ \dfrac{x-a}{b-a}, & a \le x \le b, \\ 1, & x > b. \end{cases}}$$

Equal-width intervals Δx within $[a, b]$ have equal probability:



Section 4.2 - Expected Values of Continuous Random Variables

For a function h, the expected value is

$$\boxed{E[h(X)] = \int_{-\infty}^{\infty} h(x)p(x)\,dx}.$$

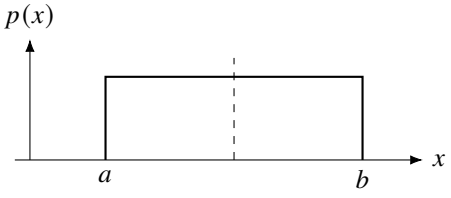
Important cases (when the relevant expectations exist) are

$$\boxed{\mu = E[X] = \int_{-\infty}^{\infty} xp(x)\,dx}, \qquad \boxed{\text{sd}[X] = \sqrt{E[X^2] - (E[X])^2}},$$

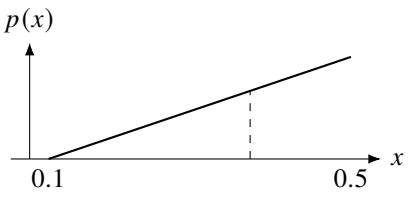
$$\boxed{E[aX + b] = aE[X] + b}, \qquad \boxed{E[X^k] = k\text{th moment of } X}.$$

If $X \sim U(a, b)$, then $\boxed{E[X] = (a + b)/2}$. For the lead-concentration model,

$$\boxed{E[X] = \frac{11}{30} \approx 0.3667}.$$



$\mu = (a + b)/2$



$\mu = 11/30$

Moment-Generating Function

The moment-generating function is

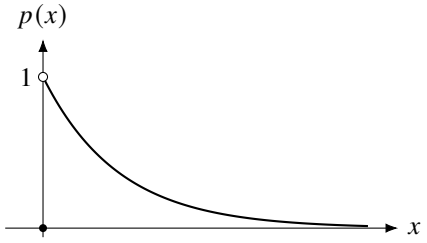
$$\boxed{m_X(t) = E[e^{tX}]}, \qquad \boxed{m_X^{(k)}(0) = E[X^k]}.$$

The derivative relation holds when the mgf exists on an open interval containing 0. Recall

$$e^a = 1 + a + \frac{a^2}{2!} + \frac{a^3}{3!} + \cdots.$$

Example 4.2.2: An Exponential-Shaped pdf

$$\boxed{p(x) = \begin{cases} e^{-x}, & x > 0, \\ 0, & \text{otherwise.} \end{cases}}$$



(a) Show that p is a valid pdf.

$$p(x) \geq 0 \quad \forall x,$$

$$\int_{-\infty}^{\infty} p(x)\,dx = \int_{-\infty}^{0} 0\,dx + \int_{0}^{\infty} e^{-x}\,dx = 0 + [-e^{-x}]_0^{\infty}$$
$$= 0 - (-1) = \boxed{1}.$$

The chain rule gives

$$\frac{d}{dx}e^{f(x)} = e^{f(x)}f'(x), \quad \frac{d}{dx}e^{-x} = -e^{-x}, \quad \frac{d}{dx}(-e^{-x}) = e^{-x}.$$

(b) Find the mean $E[X]$.

$$E[X] = \int_{-\infty}^{\infty} xp(x)\,dx = \int_{0}^{\infty} xe^{-x}\,dx.$$

Integration by parts uses

$$\boxed{\int f'(x)g(x)\,dx = f(x)g(x) - \int f(x)g'(x)\,dx}.$$

Choose $f'(x) = e^{-x}$ and $g(x) = x$. Then

$$f(x) = -e^{-x}, \quad g'(x) = 1.$$

Therefore

$$\begin{aligned} E[X] &= [-xe^{-x}]_0^{\infty} + \int_0^{\infty} e^{-x}\,dx \\ &= \lim_{x\to\infty} \frac{-x}{e^x} - 0 + 1 \\ &= 0 + 1 = \boxed{1}. \end{aligned}$$

For an indeterminate ratio of type $0/0$ or ∞/∞, L'Hôpital's rule gives

$$\lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'(x)}{g'(x)}$$

when the rule's hypotheses hold. Here,

$$\lim_{x\to\infty} \frac{-x}{e^x} = \lim_{x\to\infty} \frac{-1}{e^x} = 0.$$

Choosing $f'(x) = x$ instead leads to a harder integral involving x^2e^{-x}.

Homework

$$\boxed{2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 24}.$$

Section 4.3 - Gamma, Exponential, and Chi-Squared Distributions

I. The gamma function

II. The gamma distribution

III. The exponential distribution

IV. The chi-squared distribution

I. The Gamma Function

For $\alpha > 0$, the gamma function is

$$\boxed{\Gamma(\alpha) = \int_0^{\infty} x^{\alpha-1}e^{-x}\,dx}.$$

Gamma-Function Properties

$$\boxed{\Gamma(1) = 1}, \qquad \boxed{\Gamma(\alpha) = (\alpha-1)\Gamma(\alpha-1) \quad (\alpha > 1)},$$

$$\boxed{\Gamma(n) = (n-1)!}, \quad n = 1, 2, 3, \ldots.$$

Proof of $\Gamma(1) = 1$:

$$\Gamma(1) = \int_0^\infty x^0 e^{-x}\, dx = \int_0^\infty e^{-x}\, dx = 1.$$

Proof of the recurrence ($\alpha > 1$):

$$\begin{aligned}\Gamma(\alpha) &= \int_0^\infty x^{\alpha-1} e^{-x}\, dx \\ &= \left[-x^{\alpha-1}e^{-x}\right]_0^\infty + (\alpha-1)\int_0^\infty x^{\alpha-2}e^{-x}\, dx \\ &= \boxed{(\alpha-1)\Gamma(\alpha-1)}.\end{aligned}$$

For $\alpha > 1$, the boundary terms vanish:

$$\lim_{x\to\infty} \frac{-x^{\alpha-1}}{e^x} = 0, \qquad \lim_{x\downarrow 0} -x^{\alpha-1}e^{-x} = 0.$$

Factorial Examples

$$\begin{aligned}&\Gamma(1) = 1 = 0!,\\ &\Gamma(2) = 1\cdot\Gamma(1) = 1 = 1!,\\ &\Gamma(3) = 2\Gamma(2) = 2\cdot 1 = 2!,\\ &\Gamma(4) = 3\Gamma(3) = 3\cdot 2! = 3!,\\ &\Gamma(n) = (n-1)!.\end{aligned}$$

Example 4.2.2 Using the Gamma Function

$$\int_0^\infty x e^{-x}\, dx = \Gamma(2) = (2-1)! = 1! = \boxed{1}.$$

II. The Gamma Distribution

X has a gamma distribution with parameters $\alpha > 0$ and $\beta > 0$ if

$$\boxed{X \sim \Gamma(\alpha, \beta)}, \qquad \boxed{p(x) = \begin{cases} \dfrac{x^{\alpha-1}e^{-x/\beta}}{\beta^\alpha \Gamma(\alpha)}, & x > 0, \\ 0, & x \le 0. \end{cases}}$$

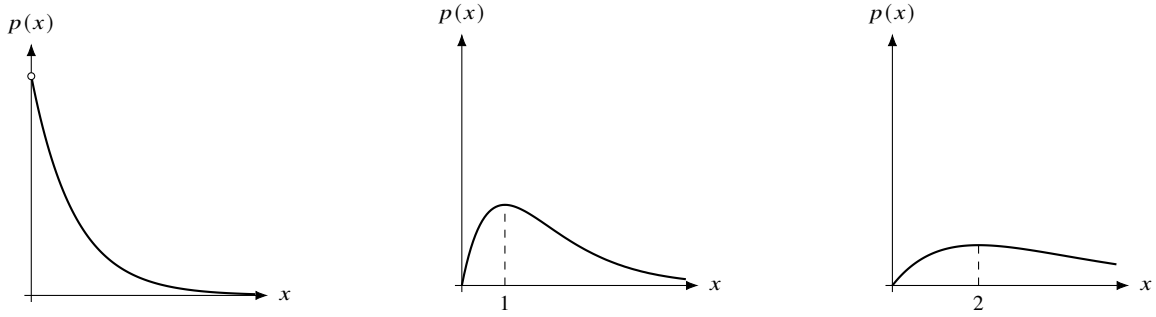
Here α is the shape parameter and β is the scale parameter.

Gamma Mean and Standard Deviation

For $X \sim \Gamma(\alpha, \beta)$,

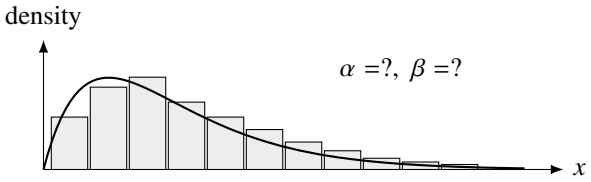
$$\boxed{E[X] = \alpha\beta}, \qquad \boxed{\mathrm{sd}[X] = \beta\sqrt{\alpha}}.$$

Three examples are $\Gamma(1,1)$, $\Gamma(2,1)$, and $\Gamma(2,2)$:



$\Gamma(1,1)$ $\Gamma(2,1)$ $\Gamma(2,2)$

For $\alpha > 1$, the mode is at $\boxed{x = (\alpha - 1)\beta}$. The graphs are positively (right) skewed.
The parameters α, β can be estimated from a relative-frequency histogram of data $x_1, \ldots, x_n$ (Chapter 7):

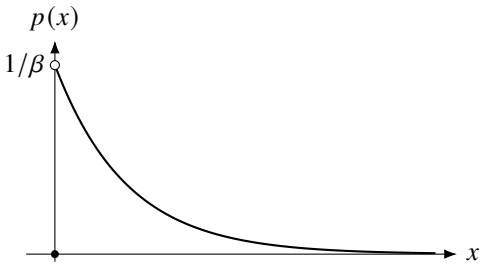


III. The Exponential Distribution

For $\alpha = 1$, the gamma distribution becomes an exponential distribution:

$$X \sim \Gamma(1, \beta) \iff \boxed{X \sim E(\beta)}.$$

$$\boxed{p(x) = \begin{cases} \dfrac{1}{\beta} e^{-x/\beta}, & x > 0, \\ 0, & x \le 0, \end{cases}} \qquad \boxed{E[X] = \beta, \quad \mathrm{sd}[X] = \beta}.$$



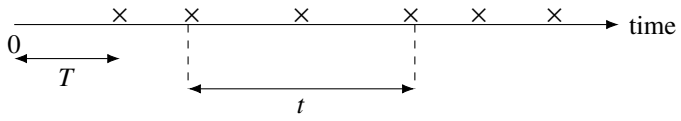
Poisson Processes: Waiting Times and Counts

For a Poisson process with rate α events per time unit,

$$T = \text{time until the first event} \quad \Longrightarrow \quad \boxed{T \sim E\left(\frac{1}{\alpha}\right)}.$$

If X is the number of events during an interval of length t,

$$\boxed{X \sim P(\alpha t)}, \qquad \boxed{E[X] = \lambda = \alpha t}.$$



IV. The Chi-Squared Distribution

The chi-squared distribution is the gamma special case with scale $\beta = 2$ and shape $\alpha = \nu/2$:

$$\boxed{X \sim \Gamma\left(\frac{\nu}{2}, 2\right)} \quad \Longleftrightarrow \quad \boxed{X \sim \chi^2_\nu}.$$

The parameter ν is the number of degrees of freedom (see Section 8.1).

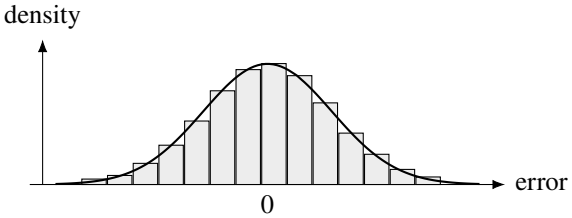
Exercises and Distribution Summary

Exercises: 32, 34, 36. For Exercises 34 and 36, assume a Poisson process.

Discrete distributions	Continuous distributions
Uniform U	Uniform U
Geometric G	Gamma Γ
Binomial B	Exponential E
Negative binomial NB	Chi-squared χ^2
Hypergeometric HG	
Poisson P	

Section 4.4 - The Normal Distribution (Gaussian)

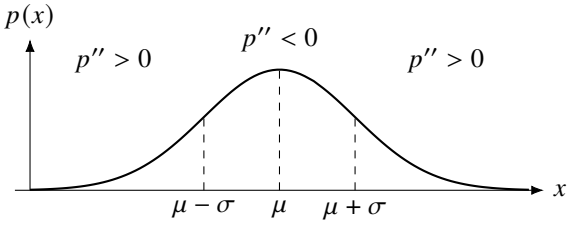
The normal distribution has a bell-shaped density. Gauss used it to model measurement errors in astronomy.



Properties of a Normal Density

The normal pdf $p(x)$ has the following properties:

1. $p(x) > 0$ for all $x \in \mathbb{R}$.
2. $\displaystyle\int_{-\infty}^{\infty} p(x)\,dx = 1.$
3. The curve is bell-shaped and centered at μ.
4. $\mu = E[X] = \displaystyle\int_{-\infty}^{\infty} xp(x)\,dx$, and
$$\sigma = \mathrm{sd}[X] = \sqrt{\int_{-\infty}^{\infty} (x-\mu)^2 p(x)\,dx}.$$
5. Its inflection points are $x = \mu - \sigma$ and $x = \mu + \sigma$.



Normal pdf and Notation

If $X \sim N(\mu, \sigma^2)$, then

$$\boxed{p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right]}, \quad -\infty < x < \infty.$$

Here $\mu \in \mathbb{R}$ and $\sigma > 0$.

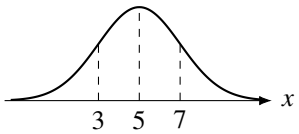
Example: $X \sim N(5, 4)$

$$\mu = 5, \qquad \sigma = \sqrt{4} = 2, \qquad \mu - \sigma = 3, \quad \mu + \sigma = 7.$$

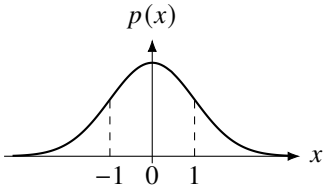
The Standard Normal Distribution

A standard normal random variable satisfies

$$\boxed{X \sim N(0,1)}, \qquad \boxed{p(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}}, \quad -\infty < x < \infty.$$



$N(5, 4)$



$N(0, 1)$